

ROS2 Architecture for Articulated Object Manipulation Using Trajectory Primitives Learned in Simulation

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Abstract The work presented in this document focuses on the ROS2 integration of expert trajectories learned in simulation. It enables robotic manipulation of articulated objects in an open environment. The proposed architecture relies on two complementary methodologies. First, a preliminary offline learning module generates trajectory primitives using the Genesis physics simulator [1]. Then, an online module selects and executes the optimal simulated trajectory to solve a given task in a specific real-world scenario. This document also includes the results and observations from the experimental deployment of this approach. The project web page, containing the code and documentation, is available at the following address: https://kappel.web.isir.upmc.fr/trajectory_primitive_website/ [2].

I. PROBLEM DEFINITION

An articulated object is an assembly of several rigid parts connected by mechanical joints (for example, for a microwave oven, the base and the door are connected by a hinge). Recent machine learning work proposes direct approaches, such as VLA methods or reinforcement learning (RL) [3], to manipulate this type of object. However, these methods are difficult to explain, which complicates the transfer from simulation to the real robot.

To overcome these limitations, we propose to learn repertoires of trajectory primitives in simulation, using the Genesis physics simulator. The ROS2 architecture then facilitates deployment on the real robot by merging the simulated expert trajectories with the real scenario. `move_group_interface` is used to plan the deployment of the simulated trajectories on the real robot.

II. VIRTUAL SCENE CONSTRUCTION

The real-world data and the trajectories learned in simulation are merged within a single MoveIt planning scene (Fig. 1). This virtual planning scene can be regarded as a digital twin, since it provides the interface between the Genesis simulation and the real world.

A. Simulated Expert Trajectories

The proposed learning module relies on the Quality-Diversity (QD) algorithm to generate, in simulation, several *trajectory primitives* for a given *action primitive*.

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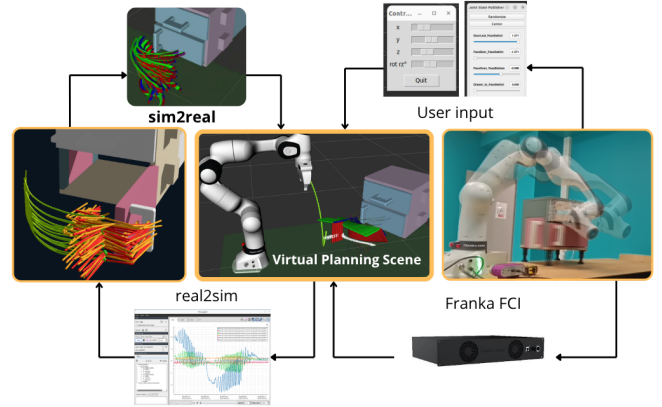


Fig. 1. **ROS2 communication between real data and simulated data.** A MoveIt planning scene combines the real configuration of the scene and the expert trajectories learned in simulation.

- *Action primitive (input)*: a task consisting in actuating a specific joint on a given object (for example, activating the hinge connecting the base and the door of a microwave).
- *Archive of primitive trajectories (output)*: a set of trajectories for manipulating a joint. All the trajectories in the result archive are sequences of waypoints of the robot's end effector in the object frame. The result archives are stored in a vector of 100 waypoints.

The GPU-parallelized learning phase provides a set of ready-to-use trajectories. To deploy these expert trajectories in the real world, a ROS2 node sends them into the MoveIt virtual planning scene. ROS2 facilitates the integration of expert knowledge into the virtual scene thanks to its tf2 tree structure, which simplifies changes of reference frame.

B. Real Robot Configuration

The virtual scene takes into account the real configuration of the robot and of its environment.

Regarding the robot configuration, information on the current state of the FR3 hardware is sent to the MoveIt virtual planning scene via a dedicated interface. In addition, another node handles the acquisition of the geometric parameters of the real environment. The permanent geometries (collision plane and safety wall) are added automatically. At the start of each scenario, the user enters, through a graphical interface, the object identifier, its 6-DoF pose and the configuration of

its joints, which allows it to be integrated into the virtual scene.

In summary, the virtual planning environment integrates both the expert knowledge learned in simulation and that coming from the real environment. The virtual planning environment bridges the gap between the simulated expert trajectories and the real scenario.

III. TRAJECTORY SELECTION AND EXECUTION

The architecture presented in this document uses MoveGroupInterface to plan and execute the robot's movements in the real world (Fig. 2). Concretely, an orchestrator node assesses the feasibility of each simulated trajectory in the real collision scenario. The first deployable trajectory, among all the simulated ones, is selected, visualized in RViz and executed on the real robot.

A. Trajectory Selection

The orchestrator node computes a plan for each learned trajectory primitive. To compute a plan, three sub-plans are computed successively in the virtual scene using OMPL.

- *Go to the pre-grasp pose*: The initial state of the plan is set to the current state of the robot. MoveIt computes the plan to reach the end effector's pre-grasp position. This frame is located 10 cm above the grasp pose itself (which corresponds to the first point of the expert trajectory).
- *Go to the grasp pose*: The initial state of the plan is set to the last virtual configuration of the robot. MoveIt computes a Cartesian path from the pre-grasp position to the grasp position on the object, following a linear trajectory.
- *Follow the activation trajectory*: The initial state of the plan is set to the last virtual configuration of the robot. MoveIt computes the Cartesian path that follows the expert activation trajectory, from the grasp point to the last waypoint.

If the chain of sub-trajectories is successfully computed in the virtual scene, the trajectory is retained for the real scenario; otherwise, another trajectory is tested.

B. Execution of the Suitable Trajectory

The proposed architecture uses the fr3 ROS2 FollowJointTrajectory controllers to execute the MoveIt plans in the real world. To make the robot more compliant, the P and D gains are modified in the fr3_controller.yaml file of the franka_fr3_moveit_config package. Using compliant control on the robot's joints allows a softer deployment and makes the robot adaptive to unexpected changes related to the simulation-to-reality transfer.

IV. RESULTS AND OBSERVATIONS

Autonomous manipulation of an experimental articulated object (modeled using Onshape-to-Robot [4]) was implemented on an fr3 in a real scenario.

The diversity of the trajectory primitives learned in simulation allowed the robot to adapt to the different real

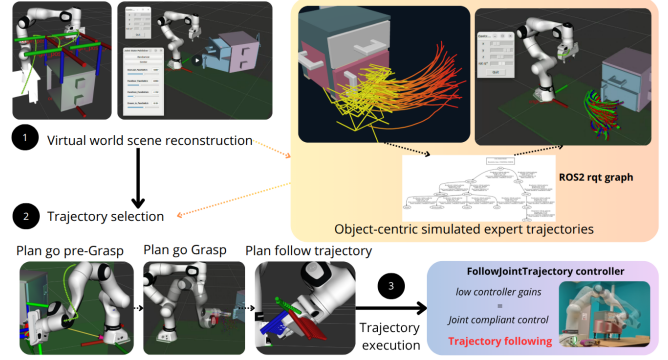


Fig. 2. **Deployment steps of a trajectory primitive.** The virtual scene is reconstructed from the real configuration and the data learned in simulation (step 1). A plan to follow the optimal trajectory is computed using MoveIt (step 2). This plan is executed by a compliant controller on the real robot (step 3).

configurations tested, thereby reducing planning failures. The MoveIt planning tolerance proved decisive for the trajectory selection step.

Moreover, during deployment on the real robot, the use of the compliant controller facilitated trajectory following by absorbing the disturbances caused by real interactions with the articulated object.

A series of deployments demonstrated the effectiveness of the method presented in this document and, through the analysis of failure cases, identified avenues for improvement. In particular, integrating torque or force feedback at the end effector would make it possible to better handle contact interfaces, this information then being able to feed the learning module for a continuous sim2real2sim loop.

REFERENCES

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